Meeting reports

Kickoff presentation

Check the kickoff presentation for context

Todo

- research all communication interfaces available for the drone
- research navigational algorithms
- research available decks for the crazyflie
- research swarm navigation algorithms inspired by nature (bee colony, ant pheromones)
- research drone 2 drone communication
- research drone rotation to scan the environment

Client demands

- the drones need spatial awareness using visual input from the ai deck camera
- the drones need to detect people and navigational elements (door, window, stairs)
- test the possibilities with the crazyflies
  - how much swarm intelligence is possible, fall back on a central compute and radio communication
- redundancy is important, the system should work for any number of drones

Week 01

Progress

- research all communication interfaces available for the drone
  - p2p radio
  - drone to central compute radio
  - ai deck to central compute image streaming
- research available decks for the crazyflie
  - positioning nodes/decks are not viable due to power and complexity issues
  - flow deck and ai deck are a viable combination for position aware flight
  - the flow deck cannot be used to capture eagle eye camera shots
- research drone 2 drone communication
  - wifi and bluetooth are not viable
  - p2p communication is viable via radio
- research drone rotation to scan the environment
  - this is a viable option using gaussian splatting algorithms to convert 2D images to a 3D environment

Todo

(deliverables)

- single non flying drone image stream to central server, existing object detection on central server
- single flying drone image stream to central server, map making on central server
- single non flying drone image stream to central server, position inference on central server
- make telemetry report of flying drone from origin, moves, to origin (check kalman)
- further experiments on P2P with 2 or 2+ drones
- interface that shows the drone path

client feedback

- use a wooden pole, black and white pattern as a navigational element
- research what are hardware options for inference on the drone itself

Week 02

Progress

- there is an api where you can upload images captured from the drone, these images get stitched together to form a pointcloud
- we have a router to create a network wifi network to stream these images, every drone gets a static ip address
- there is progress on drone collision avoidance using a P2P radio connection using a hardcoded predetermined position for each drone
- the flow deck is viable even on a sloped surface, we discovered this after testing in the old swimming pool
- the flow deck is can get confused by floor texture, we discovered this by testing in our working environment
- small deviations from course can result in a large accumulative error
- The gap between drones for collision avoidance should be bigger, 0.5m is not enough
- right now using drone net for detecting obstacles, it is an optimized model but for older hardware
- drone to environment collision avoidance appears difficult because the repositories are outdated
- yolo26n retrained with labeled images taken with the ai deck camera
- lighting can mess with the camera

Todo

- create a pointcloud map from the images of multiple drones
- we need to think on how to integrate all the different sensors and systems (ai deck camera, flow deck, positioning, mapping)
- collision avoidance using a P2P radio connection needs to be improved and extended to more then 2 drones
- combining ai deck map navigation position inference and flow deck position estimate
- look into whole application architecture also including pre run necessities and communicating systems, delays during runs
- explore multiranger deck possibilities
- for drone to environment collision avoidance we need an environment that has more obstacles
- look into compression techniques for models for drone to environment collision avoidance and mapping

- try using segmentation or depth models for drone to environment collision avoidance
- need a way to remember previously detected items

Client feedback

- track progress with incremental steps, focus on documentation
- detect drones and know which drone is being detected, know your nearest neighbour
- use homeing beacons (an object to recognize for the ai deck), this is redundant because of the mapping system
- speed is not important for this project, drones can go slowly
- document what would be needed for on board processing to detect obstacles (try raspberry pi)

Extra

- we now have the ranger deck available

Week 03

Progress

- mapping works perfectly for a single drone with an api image upload
- image processing for map making takes 30 seconds per image
- each drone is granted a static ip for a router network
- using set positions drones can communicate P2P for collision avoidance
- the drones can handle a sloped surface, the ranger deck does not get confused, the texture of the floor is the most important factor
- testing of the error accumulation of the ranger deck resulted in an average of 10cm after a $1m^2$ square flight sequence, but the takeoff and landing introduce even more error
- drones flying within 0.5m of eachother is dangerous for the drones due to error accumulation in the IMU
- using drone net to detect obstacles, this is not viable
- using the drone image stream we can use yolo26n for object detection
- discovered lighting has a big influence on the camera's performance

Todo

- figuring out a way to stitch together maps to create a common map (multiple drones produce multiple maps that need to be merged)
- speed up image processing for map making
- infering position from drone image maps and a drone image
- getting started with the multi ranger deck for room mapping
- P2P collision avoidance with 3 drones
- creating a test environment for collision avoidance
- use depth models on the drone itself, try pulp tiny v3

Client Feedback

- focus on documentation, track progress with incremental steps
- try to detect drones and know which drone is being detected, know your nearest neighbour

- use homeing beacons, an object to recognizere (this is redundant because of mapping)
- determine what would be needed for onboard object detection models
- look into the whole application architecture: pre run necessities, communicating systems, delays during runs
- remember previously detected items

Week 04

Progress

- state claasification implemented but performs badly due to drone image quality (it is a retrained mobilenet model)
- mapping different frames from different cameras is still difficult but works, in documentation all solutions have a camera that can detect depth, problem with combining drone relative position is depth estimation is not alligned
- ranger deck first, ai decks second approach can perform wall following with ranger decks and mapping safe zones, objects of interest and ai decks fly afterward
- ranger deck, ai decks grouped approach is difficult because of undetected objects that the following ai decks can crash into and needs more testing and a central compute system because of the limitted compute on the drone (192 kilo bytes)
- all drone research is stumped because of position estimation problems, limitted on board computation and the inexperience of firmware programming and lack of knowledge in navigational algorithms, a simulated testing environment could remove a lot of these drawbacks

Todo

- we will try models where the drone position is part of the input parameters
- improving ranger deck first, ai decks second approach
- improving ranger deck, ai decks grouped approach
- finalise deliverables
  - visual demonstration of the ranger deck, ai decks grouped approach
  - integrate ranger deck first, ai decks second approach with image streaming and processing (object detection and pointcloud)

Client feedback

- look into a human operated swarm, setting waypoints for drones to go to
- try using two radio's (this is mathematically impossible you need three)

Week 05

Todo

- demo filming
- state estimation model improvement
- documentation of successfull and failed approaches
- finalising all project management deliverables

Meeting notes

Week 01

check client kickoff presentation

Week 02

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360 video

communicate position to central server

only fly forward with turning

minimal viable product,

any object detection

deliverables week 1

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deliverables, week 2
- single non flying drone image stream to central server, existing object
detection on central server
- single flying drone image stream to central server, map making on central
server
- single non flying drone image stream to central server, position
inference on central server
- make telemetry report of flying drone from origin, moves, to origin
(check kalman)
- further experiments on P2P with 2 or 2+ drones
- interface that shows the drone path

further possiblilities

- wooden pole, black and white pattern as a navigational element
- what are hardware options for inference on the drone itself, what are the
requirements (research)

- p2p navigational algorithm
- create a drone mapping interface

- create a control drone visual interface


### project board

everyone takes one hour, no coding to improve their issues
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Week 03

```
mapping works perfectly, api image upload
stitching together drone maps, to create a common map
- create map using multiple drones
each drone sends images using wifi
each drone is granted a static ip
drone collision avoidance, using P2P radio, with set positions
sloped surface is viable, texture is important (immersive room or swimming pool)
working now to intergrate different sensors
testing using simple drone with square sequence run 10% error with accumulation
- P2P collision avoidance with > 3 drones
track progress with incremental steps, focus on documentation (remark from Bart)
nikita encountered accumulation between the drones 0.5 meters starting distance between drones is not enough
- combining ai deck map navigation position inference, and flow deck position estimate
30 seconds per image processing for map navigation
- (bart) detect drones and know which drone is being detected, know your nearest neighbour
- use homeing beacons (an object to recognize) -> redundant because of mapping positioning is underway
obstacle avoidance, problem with testing environment, not enough walls, creating cardboard environment or pillows, (bart) go slowly to avoid collision
- (bart) what would be needed for on board processing to detect obstacles (try raspberry pi)
right now using drone net for detecting obstacles, it is an optimized model but for older hardware
(for roel) look into number of parameters and compression techniques
(personal idea) using segmentation, (tofa idea) using depth models, on the drone ai deck itself
pulp tiny v3 ... but it seems the most logical choice
- look into whole application architecture also including pre run necessities and communicating systems, delays during runs
we received a multiranger deck, use it as a single example
we can detect people using yolo26n on the server, nazar improved with retraining yolo26n on the server
lighting is a big edge case for the camera
next detection step, don't forget about previously detected items, abandon navigational item detection
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interim

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A drone can scan anythin
A swarm explores faster, there is a need for adaptive behaviour
current solution, master slave, ranger + follower ai deck
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let ranger deck drone swarm around obstacles, create a map of environment
with a large padding around obstacles to protect ai deck drones
space must be unknown
don't use deterministic models
we have access to multiple ranger decks

- explore 2 approaches
    - drone groups consisting of ranger deck and ai decks
    - ranger deck swarm first, ai deck swarm second

first only few obstacles
look into obstacle wall follower

future research, project ai deck images on ranger deck map

deliverables
    - opportunities
    - explored options
    - what detection models are runnable on ai deck
    - central ranger deck mapping
    - extra decentralized ranger deck mapping
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Week 04

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last time the model recognized people
state classification is already implemented
using yolo26n retrained and mobilenet retrained running on the central
computer

mapping different frames from different cameras is still difficult but
works
in documentation all solutions have a camera that can detect depth
we will try models where the drone position is part of the input parameters
problem with combining drone relative position is depth estimation is not
alligned

drone wall following with ranger decks and mapping safe zones, objects of
interest and ai decks afterward

human operated swarm, setting waypoints for the drone

trying master slave drone communication
central compute system is requireed because of lack of computation
1 in front, 2 in the back

master slave relation is difficult because of undetected objects that the
following ai decks can follow

relative positioning using flow deck only is not used
using two radio receives you can get positioning using the angle of
reception
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all teams struggled with position estimation, limitted compution,
difficulty of firmware programming

a virtual environment can remove a lot of limitations our drones experience

- visual demonstration of the master slave approach
- integrate, ranger deck - ai deck approach, object detection of multiple
drones, send image data to pointcloud server
    - annotationg region of interest with ai deck detection seems a bit
difficult
    - focus more on random exploring
    - explain pros and cons
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Week 05

```
focuss on deliverables and documentation
we will focuss on deliverables this week unfinished reasearch (ranger deck,
ai decks group / improving the person state estimation model)
film systems (ranger deck first, ai decks second / image streaming, object
detection, map making)
also mention failed approaches
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